

Artificial intelligence in allergy and immunology: Recent developments, implementation challenges, and the road toward clinical impact



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Artificial intelligence (AI) is increasingly recognized for its capacity to transform medicine. While publications applying AI in allergy and immunology have increased in number, clinical implementation substantially lags behind other specialties. By mid-2024, over 1,000 US Food and Drug Administration–approved AI-enabled medical devices existed, but none specifically addressed allergy and immunology. This gap partly reflects the field’s limited reliance on imaging, which facilitated early AI breakthroughs in radiology and pathology. This narrative review examines recent AI developments, including large language models and AI agents, evaluating their applicability to allergy and immunology practice. We analyze current and potential applications, emphasizing those demonstrating clinical value while identifying implementation barriers amplified by allergic diseases’ unique complexities, including data privacy concerns, bias, reliability constraints, and evolving regulatory frameworks. To bridge the persistent research-to-implementation gap, we propose a 6-point road map: (1) prioritize impactful applications, (2) define clinically relevant benchmarks, (3) enforce rigorous governance, (4) transition to operationalization, (5) promote clinical adoption through trustworthy AI, and (6) establish life cycle management. This road map builds on established implementation frameworks while incorporating critical

field-specific considerations unique to allergy and immunology. Through this approach, we provide a perspective for advancing AI in allergy and immunology from academic promise to tangible clinical benefit. (*J Allergy Clin Immunol* 2026;157:295-306.)

Key words: Large language models, AI agents, clinical adoption, implementation barriers, validation frameworks, trustworthy AI, AI governance

Artificial intelligence (AI) is increasingly recognized for its capacity to transform the field of medicine.¹⁻³ AI is a subfield of computer science that can simulate human intelligence and perform tasks that previously required humans. In the field of allergy and immunology, AI has already demonstrated many applications, including predicting asthma by analyzing the text of electronic health records,⁴ diagnosing allergy from epigenetic data,⁵ and identifying eczema from image data.⁶

The field of AI is developing rapidly, with most recent advancements based on increasingly large deep neural networks.^{7,8} Recent breakthroughs in large language models (LLMs) and multimodal AI form the methodologic basis for many new applications and the surge of generative AI.⁸⁻¹¹ ChatGPT,¹² which generates text and images from user prompts,¹³ exemplifies these advances. Despite a major increase of AI publications⁷ in medicine—nearly 60,000 were listed in 2024 in PubMed—large-scale translation to clinical implementation in allergy and immunology remains a significant challenge. For example, by mid-2024, there were slightly more than 1,000 US Food and Drug Administration (FDA)-approved AI-enabled medical devices,^{14,15} yet none was related to the field of allergy and immunology.

Why is allergy and immunology lagging compared to other specialties? First, the initial wave of (FDA approved) AI applications predominantly focused on medical imaging,¹⁶ thereby advantaging specialties with digital imaging workflows such as radiology and pathology. Second, allergy and immunology rely less on image data and more on clinical and laboratory data, which have fewer AI-driven, FDA-approved use cases. This may have resulted in lower AI literacy and uptake in the field of allergy and immunology. More importantly, this gap highlights critical questions about what our field needs to catch up with and what realistic expectations we should have for AI implementation.

The objective of this narrative review is to provide an expert perspective on these questions. We aim to: (1) critically analyze recent AI developments (2022-25) and their applicability to allergy and immunology, (2) identify implementation barriers,

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and (3) propose a road map for translating AI research into clinical practice.

A general introduction to the basic concepts of AI is provided elsewhere, including various excellent reviews, some with a specific focus on allergy and immunology.^{3,13,17-21} To further introduce AI concepts, Fig 1 provides an introduction to current AI methods and their applications in daily life and medicine.

NOVEL DEVELOPMENTS IN AI

Recent advances in AI encompass 3 major developments: LLMs,^{2,11,22,23} multimodal learning, and AI agents. Each of these is addressed in turn. Definitions are listed in Table I. These developments within (generative) AI present novel research opportunities and clinical applications while also raising important questions regarding responsible and trustworthy use.

LLMs

LLMs have shown unprecedented capability to both interpret text data and generate human-level text. This makes them suitable for a wide range of applications. For example, although previously challenging to analyze, LLMs have been applied to electronic health record processing with efficiency comparable to structured data analysis.²⁴ This has unlocked advancements in patient identification in the coronavirus disease 2019 pandemic,²⁵ clinical trial matching,²⁶⁻²⁸ prediction of disease progression,²⁹ and readmission risk.³⁰ In neurology, LLMs have provided novel insight into disease trajectories and early detection of neurodegenerative disorders through clinical documentation analysis.^{31,32} Moreover, while LLM have been developed to handle text, other medical data, such as DNA sequence information, can also be seen as text and analyzed with LLM models.³³ For example, LLMs can be used to understand the functional impact of genetic variations, and in proteomics, they can be used to predict the 3D structures of proteins.

In recent years, LLMs' performance³⁴ has surged, together with commoditization of these models³⁵ and their open-source availability.³⁶ Recently, strong improvements in efficiency and cost-efficient use of LLMs were obtained with the release of DeepSeek V3 and R1.^{37,38} This allows researchers to widely utilize LLMs without having to train their models, which would be impractical as a result of the enormous size of state-of-the-art LLMs, which have billions of parameters as well as high complexity. Consequently, many LLM applications make use of publicly available pretrained foundation models from established providers, adapted for downstream tasks.³⁹

From a practical implementation perspective, composing prompts—the structured instructions provided to the LLM—constitutes the most critical determinant of model performance. LLMs typically process two distinct textual inputs: the system prompt, which establishes the foundational behavioral parameters and operational context (often predefined and invisible to end users), and the user prompt, which conveys specific task instructions. Fig 2, A and B, illustrates the essential elements of prompting and the underlying mechanics of LLMs via a patient education scenario.

The output of an LLM depends critically on the prompt engineering methodology. Chain-of-thought prompting, whereby LLMs are guided to explicitly perform sequential reasoning steps, represents one of the most effective methodologies for enhancing model performance.⁴⁰ Thus far, the current academic literature

often lacks rigorous prompt-optimization protocols,^{41,42} directly affecting output quality.^{43,44} Selection and validation of prompts warrant equivalent scrutiny to statistical model variable selection, and researchers should report about this systematically and transparently.⁴⁰

Multimodal learning

Multimodal learning is the combined use of diverse data modalities in a single application.^{45,46} Contemporary foundation models, including Gemini 2.0 and GPT-4o, inherently process multiple data types. This capability enables combined analysis of previously disparate medical data sources, such as integrated -omics and imaging data, or clinical documentation with telemonitoring data. In clinical settings, the rise of AI medical scribes is an important use case of multimodality, where speech is transcribed and summarized during a consultation. It has been shown that these applications offer the potential to reduce the administrative load of health care professionals.⁴⁷

Most model providers have recently released reasoning models—that is, large models specifically optimized for complex tasks requiring multistep analytical processes, often across data modalities.^{48,49} These developments have yielded large improvements in medical knowledge assessments, with scores on the MedQA-USMLE benchmark advancing from Google's Med-PaLM at 67.6% in 2023 to OpenAI's o3-mini-high at 90.55%, approaching expert-level competency.^{50,51} However, recent research has identified significant failure modes in these large reasoning models, including brittleness under adversarial conditions and inconsistent performance across related tasks.^{52,53}

AI agents

AI agents or agentic AI represent a recent evolution beyond discrete task completion toward autonomous decision-making (Fig 2, C).⁵⁴ An *AI agent* can be defined as an AI system that is designed to perform a specific task, often involving decision-making, problem-solving, knowledge retrieval, and interaction with other systems.⁵⁵⁻⁵⁸ They thus have the potential to perform a range of tasks previously requiring human involvement. Agent networks or multiagent systems have emerged as a focus of recent research, wherein multiple AI agents collaborate (Fig 2, D) toward system objectives, mirroring the ways that humans approach objectives.⁵⁹ The rise of protocols like model context protocol, or MCP, is a key enabler of agent interconnectedness. This standardized protocol enables AI agents to collaborate and share context, much like application programming interfaces, better known as APIs, once connected traditional applications.^{60,61} MCP allows agents to dynamically exchange information and coordinate tasks, streamlining complex workflows across systems. However, health care implementation of such new protocols faces substantial barriers, including security requirements and the prevalence of fragmented legacy systems that may greatly delay adoption timelines.

Medical applications are emerging gradually, however, notably in health care organizational workflow automation.⁵⁵ Commercial entities already promise to extend health care workforces with AI agents,⁶² offloading existing staff by performing administrative tasks. The reliability and adoption of such promises remain to be seen. In parallel, AI platforms facilitate various research activities, such as literature collection and synthesis

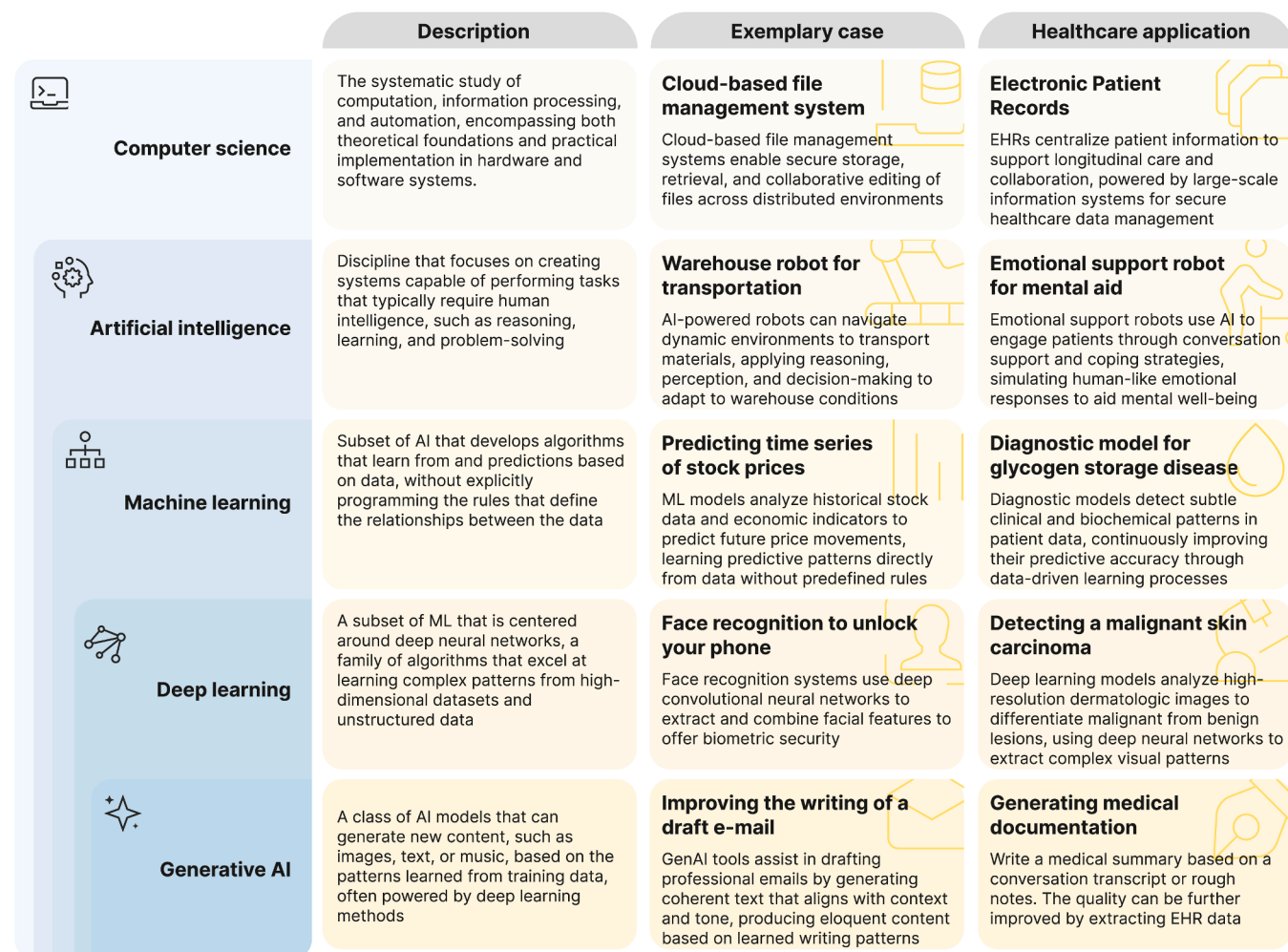


FIG 1. Overview of AI-related disciplines with illustrative cases and health care applications.

(Scite AI, Afforai, Gemini Deep Research, Consensus⁶³⁻⁶⁶), code development (Cursor, GitHub Copilot^{67,68}), and data analysis. Given their apparent consistency and eloquence (and their as-yet far-from-perfect quality), it is essential for researchers to critically assess the output of these systems. For this review, Scite AI, Afforai, and Gemini Deep Research were used to assist us with the initial literature search, and we used Claude to improve readability; none of these tools was used to generate new content.

One of the most pressing topics of the applicability of these LLM-based methods in health care is their risk of producing what are termed *hallucinations*, or incorrect and misleading responses that are positioned confidently and that often sound plausible.⁶⁹ Given the potentially catastrophic effects that this could have in clinical practice, hallucinations remain a major challenge in clinical implementation and adoption. Despite progress on detection and mitigation measures,⁸ hallucinations persist in nearly all models, leading some scholars to argue for fundamentally different architectures.⁷⁰

Another limitation of current technologies is their restricted domain-specific knowledge. While state-of-the-art models demonstrate robust performance across broad applications, they may exhibit suboptimal functionality in highly specialized clinical contexts. Various techniques have supported more

tailored use, such as fine-tuning and retrieval-augmented generation.⁷¹ Fine-tuning involves an additional training phase that customizes the LLM for specific applications, such as incorporating domain-specific literature focused on allergy and immunology. Conversely, retrieval-augmented generation retrieves information from external knowledge sources, such as clinical guidelines, during the generation process. This methodology enhances reliability by grounding LLM outputs in established scientific and clinical evidence.

AI APPLICATIONS IN ALLERGY AND IMMUNOLOGY

The use of AI in allergy and immunology is increasing, with potential for integration in multiple areas including diagnosis, management, prevention, self-care, and research.¹³ In theory, with the use of AI, analyses of large, unstructured, and complex datasets can be performed quickly, reveal potential patterns in the data, and help develop patient-specific models and diagnostic or management algorithms for use in clinical practice.^{18,72}

For diagnosis, machine learning could be used to analyze results of diagnostic tests (for example, food-specific IgE), combine these with clinical history, and develop algorithms that

TABLE I. Key terms and definitions in field of AI

Term (abbreviation)	Definition
AI agent	AI system designed to perform specific tasks, often involving decision-making, problem solving, knowledge retrieval, and interaction with other systems.
Convolutional neural network (CNN)	Specialized neural network architecture that processes grid-structured data (like pixels in images) by applying filters to detect spatial patterns and features.
Generative pretrained transformer (GPT)	Specific architecture of large language model developed by OpenAI that processes text through transformer neural networks.
Large language model (LLM)	Form of AI that utilizes natural language processing tasks for machine learning, often for purposes of prediction.
Large reasoning model (LRM)	Form of AI that utilizes natural language processing and structured reasoning to solve complex, multistep problems.
Natural language processing (NLP)	AI type that uses machine learning to allow computers to both understand and generate human language for purposes of communication.
Neural network	AI type by which learn using interconnected nodes intended to mimic working of human brain.
Node	Computational unit within neural network that receives inputs, applies mathematical operations, and transmits outputs to other connected nodes.
Prompt	AI input that user provide that instructs AI model to generate specific output.
Retrieval-augmented generation (RAG)	Technique that enhances AI text generation by first retrieving relevant information from external knowledge sources.

will assist physicians in reaching a more accurate diagnosis of food allergy. Some initial efforts have been reported in combining data from multiple clinical instruments and biomarkers to help predict the outcome and severity of allergic reactions during oral food challenges.⁷²⁻⁷⁴ For anaphylaxis, machine learning models may identify anaphylaxis from medical records.¹⁹ Similarly, for asthma, machine learning models may help distinguish patients with asthma from patients with other respiratory diagnoses, or they may potentially predict asthma exacerbations from risk factors.^{3,13,19} In atopic dermatitis, the use of AI may not only facilitate the diagnostic process but also help tailor treatment and improve patient outcomes thanks to the development of phenotypes facilitated by data aggregation.⁷⁵ Data mining of clinical records has also been used to facilitate diagnosis,^{75,76} although these studies suffer from relatively small, single-cohort designs that cannot make claims regarding generalizability.

The development of tools capable of interpreting images, as spearheaded more quickly and accurately in other fields, may produce a new uptick in the adoption of AI-based technologies. Several examples of this approach have been carried out. For instance, a convolutional neural network model was trained to automatically diagnose atopic dermatitis with multiphoton tomography imaging; it had sensitivity and specificity of >0.95%.⁷⁷ Moreover, high-resolution computed tomographic pulmonary images have been analyzed with machine learning to differentiate children with severe asthma from healthy children. This approach displays an overall accuracy of 90-98%⁷⁸ and measures features such as airway wall thickness, although the sample size was limited (n = 41) and not extended on prototyping. AllergoEye⁷⁹ is an AI method to interpret conjunctival allergen provocation tests from photographs of eyes after exposure to allergens (Fig 3). Although initial results showed 98% sensitivity and 90% specificity compared to human assessment, the developers have not yet demonstrated reproducibility across different populations or clinical environments—a key limitation for clinical translation.

Similarly, convolutional neural network models have been trained to detect positive reactions to multiple allergens in skin prick tests images (thermography).⁸⁰ This approach reaches an

area under the curve of 0.98, but it does not substantiate its claims regarding reduced examination time and reduced subjectivity. One concern relates to algorithmic bias, which could be addressed by training the model using patients with differences in skin color. In a recent study, asthma-related symptoms for predicting exacerbations were accurately extracted from clinical notes using transformer-based deep learning methods,^{81,82} demonstrating the potential of using language models as a powerful tool to gain new insight from existing data. However, future studies should demonstrate how this approach could be translated into clinical practice to effectively prevent exacerbations.

For disease management, one potential use of AI may be to examine and analyze treatment response, with the aim of adjusting dosages, optimizing therapy schedules, improving outcomes, and minimizing adverse reactions. After being provided with individual patient information such as history, risk factors, and biomarkers, AI may be able to predict the treatment course, thereby permitting adaptive and individualized protocols in the context of personalized medicine.¹⁸ A recent cross-field review demonstrated that AI predictive analytics improved patient outcomes (diagnosis, disease progression, treatment response, and recovery rates) through machine learning and deep learning techniques that enabled early detection and personalized treatment approaches.⁸³

Other applications of AI include its use to improve practice management, including performing a variety of administrative tasks such as automated clinical order sets and digital scribing for documentation.¹⁸ This could reduce the administrative burden of clinicians and allow them to focus more time on providing direct patient care.⁸⁴ Natural language processing can help analyze unstructured clinical patient notes, prepare reports, document patient interactions, and conduct conversations.^{3,18,20} Between 2023 and 2024 alone, a survey of 1,000 physicians noted an increase in AI use from 38% to 66%,⁸⁵ primarily for administrative work.

Moreover, LLM-powered approaches have also demonstrated potential in health literacy enhancement by delivering personalized medical information and explaining complex medical

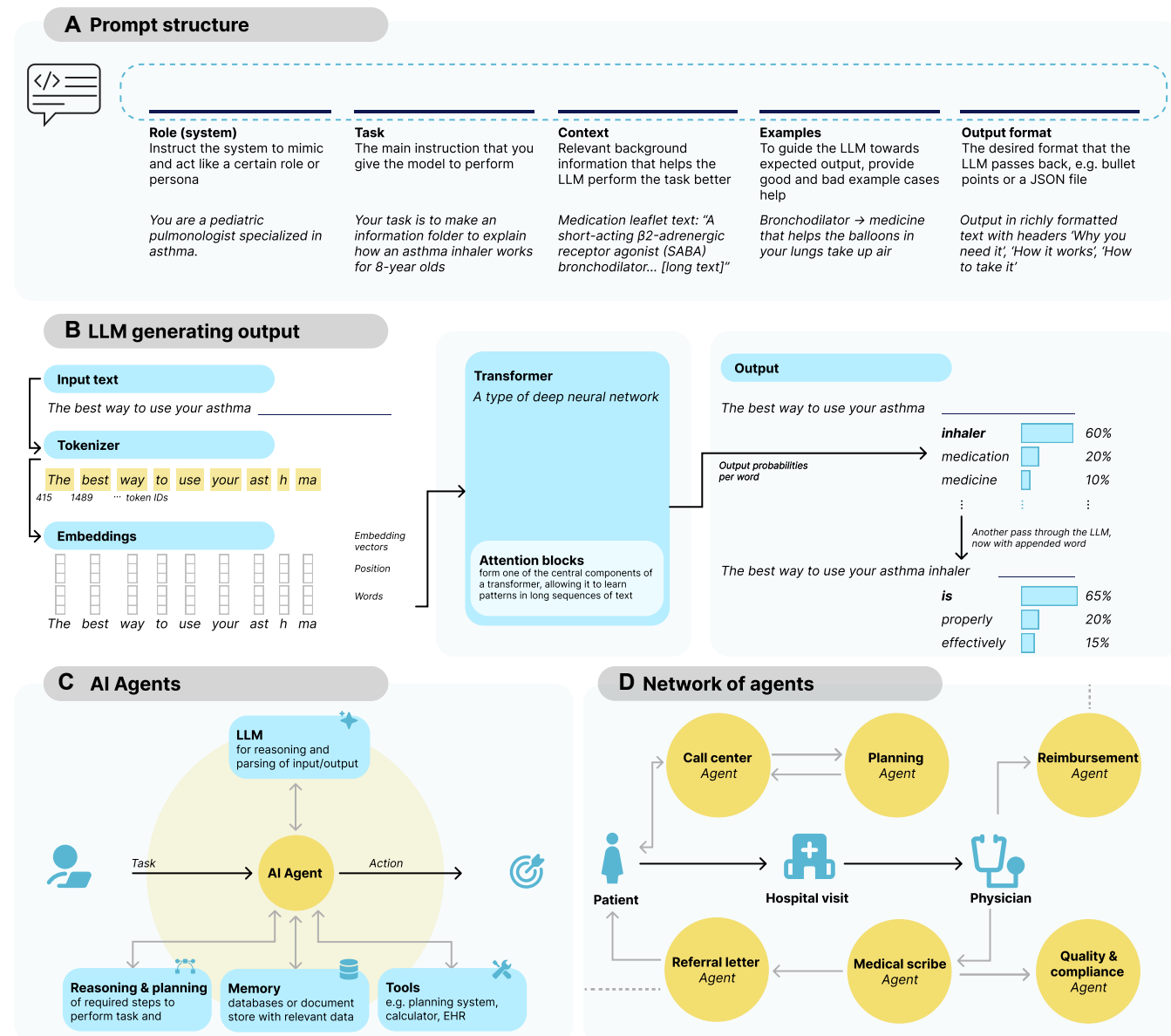


FIG 2. Schematic explanation of LLMs and AI agents. Delineated are architectural components and operational principles underlying LLMs, illustrated in context of allergy. **(A)** Prompt engineering framework illustrates best practices for creating prompts. **(B)** Generative inference elucidates how LLMs predict the next words in sequences. The process starts with tokenization, which splits words into tokens, followed by dimensional embedding, which transforms words to numerical vectors that preserve semantic and syntactic relationships. These embeddings are generally processed via a transformer-based architecture that models long-range dependencies within the input sequence, culminating in probability distribution across vocabulary space. **(C)** Agent-based architecture depicts augmentation of core LLM capabilities through integration with other modules. This enables multistep planning, patient-specific data incorporation, and interoperability with clinical information systems to perform more complex actions. **(D)** Multiagent network conceptualizes a system of interconnected AI agents that coordinate discrete functions within hospital operations. *JSON*, JavaScript Object Notation (open standard file and data interchange format to store and transport data).

terminology in someone's native language,⁸⁶⁻⁸⁹ as well as improving physicians' clinical reasoning.⁹⁰ One recent relevant example shows that ChatGPT (GPT-4.0) delivers 96.3% accuracy in providing asthma patients with health-related information, based on the patient-completed Asthma Knowledge Questionnaire.⁹¹

Furthermore, AI may enhance the monitoring of airborne allergens and its related risk assessment, which could prevent worsening of allergy symptoms. A notable example is a pollen proxy that enables pollens to be monitored over time with a combination of neural network and random forest models and affordable optical particle counters.⁹² Interestingly, this type of

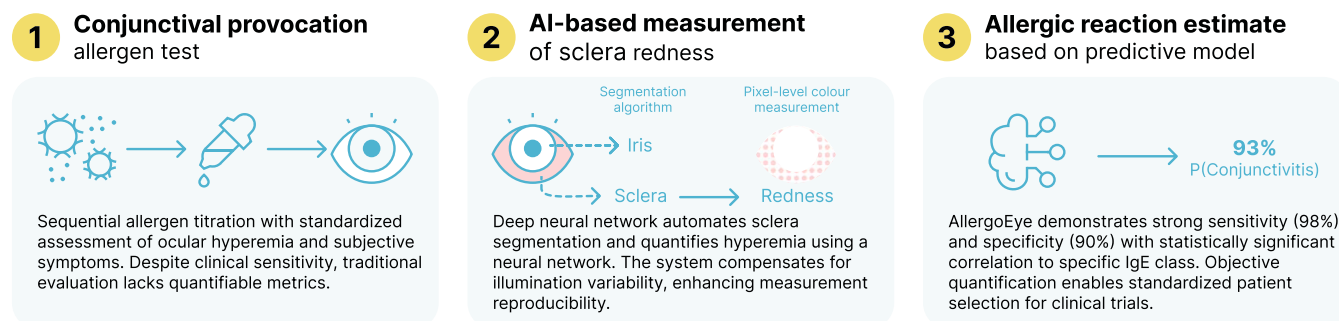


FIG 3. Application of AI-based measurements to improve conjunctival provocation tests. Approach enables standardized, quantitative interpretation of allergic responses but requires external validation in heterogeneous clinical settings before it can be widely adopted.

low-cost application could bring widespread implementation closer. Furthermore, researchers developed a portable AI label-free sensor capable of detecting pollen and mold spores with an accuracy of >94%.⁹³ Inspired by these advances, several different public health institutions have promoted initiatives to use real-time pollen data to prevent unfavorable health outcomes.⁹⁴

Finally, the self-care of allergic patients could also benefit from the AI explosion. Personal devices such as smartphones and wearables are valuable tools to extract data and feed machine learning algorithms. Various cough-counting applications have been piloted with accuracy comparable to human annotators,^{95,96} although they are not yet out of the prototype phase. Moreover, a device (known as SONU) to alleviate hay fever–driven nasal congestion uses an AI image-based algorithm to scan the user's face and predict the geometry of the nostrils and paranasal sinuses in order to optimize the functioning of the device.⁹⁷ The cost of these devices and insurance coverage should be taken into consideration before broad implementation becomes feasible.

Recent studies demonstrate AI's emerging role in research, such as -omics data analysis,^{98,99} offering complementary approaches to established biostatistical methods such as epigenome-wide association study and principal component analysis. Deep learning models can capture the intricate structures within these data, both within and across -omics layers.^{100,101} Transformers, a type of deep learning architecture originally designed for (and still heavily used in) natural language processing, have been used for -omics integration because of their ability to handle long-range sequential data.¹⁰²⁻¹⁰⁴ This could be particularly powerful in allergy research, with a highly relevant interplay between -omics layers and environmental factors.¹⁰⁵ Similarly, the deep learning architecture of autoencoders has been proposed for feature extraction within high-dimensional -omics data.^{106,107} This has been shown to improve downstream analysis, such as disease prediction or endotype clustering.¹⁰⁸ Multimodal analysis platforms, such as PandaOmics,¹⁰⁹ integrate multiomics with text-based gene and disease information to find novel therapeutic targets and biomarkers. This platform was used to identify targets (*IL4R*, *IL5*, *JAK1*, *JAK2*, *JAK3*, and *NR3C1*) for allergic diseases.¹¹⁰

Although these examples hint at the technology's promise and potential applications, most AI applications have not reached the bedside—at least not yet. This implementation gap has been termed the *AI chasm* or the *last-mile problem*.¹¹¹ The reasons for the existence of this gap are many and are beyond the field-specific reasons we mentioned earlier, including the fact that

information provided by AI (1) may not be fully comprehensive, (2) may contain inaccuracies, and (3) lacks the critical judgment that clinicians possess.⁸⁶ Additionally, from physicians' point of view, fear of a loss of professional autonomy and difficulties integrating AI into clinical workflows have been reported to be hindering factors, but receiving training on the use of AI has facilitated acceptance.¹¹²

Recent quantified estimates of the “width” of this gap are lacking, but a systematic review by van de Sande et al¹¹³ concluded that there are few randomized controlled trials with AI clinical decision-support systems and that the evaluation of these studies was too limited and incomplete to properly assess the impact on clinical practice. Nair et al¹¹⁴ found that just 38 articles met the criteria for properly evaluating the implementation approach for their AI application.

Finally, field-specific complexities of allergy and immunology further complicate implementation. First, standardized datasets are more rare, and heterogeneous datasets are needed to properly capture the environmental and population-level difference, making generalization and validation harder. Second, the transition from pediatric to adult allergy care and related age-dependent disease patterns make AI applications harder to deploy.^{21,115}

BENEFITS AND RISKS OF AI

Many technical and ethical challenges exist in AI implementation that merit careful consideration and are gaining attention.^{84,116-120} First, safeguarding the privacy of patients' sensitive data is a major consideration.^{20,120,121} Second, the accuracy of AI is heavily influenced by the quality of the data provided. For example, if the data are low quality and rife with heterogeneity, this will limit the accuracy of the diagnostic algorithms that are output. If the data are biased—for example, if they are skewed toward certain demographics and populations—this lack of diversity can create problems of disparities and inequity and can miss valuable potential trends with regard to heterogeneity of potential outcomes and effects.^{19,20} Algorithmic bias is particularly pressing in the field of allergy and immunology because of the complex environmental interactions, heterogeneous disease manifestations, and data-size limitations that characterize this discipline.

More specifically, the lack of sufficient phenotyping of individual patients is a significant challenge for the applicability of AI to clinical practice in this field. Moreover, the clinical information contained in health records is often scarce and shows a great

variability with regard to completeness among patients and among follow-up visits. These facts greatly complicate the training of AI tools and ultimately account for their low clinical implementation. Third, because AI is a new development in the medical field, wide adoption will require physician education and training so they can better understand the benefits, risks, and limitations and use its applications appropriately.^{18,122} Health care providers should be trained how to collect clinical data from patients and how to be aware of bias so that the information obtained will have sufficient quantity and quality for AI applicability. Additionally, it may take time before health care providers feel comfortable using AI and place their trust in it. At the other extreme, an overreliance on AI may reduce the critical judgment that physicians possess and lead to mistakes.¹⁸ In such situations, it would be difficult to determine who would take responsibility for this error.^{18,19} Therefore, regulatory authorities need to adopt policies on how to regulate AI applications. Experts advocate for interdisciplinary collaboration among AI developers, health care providers, and policymakers to ensure ethical and responsible AI implementation in clinical practice.¹²³⁻¹²⁵ Finally, the financial costs associated with wide implementation of AI can also be a limitation in many resource-poor settings. There is no current guidance regarding reimbursement for the use of AI, and it is unknown if its use will be cost-effective.¹⁸ Scholars have identified challenges with existing reimbursement mechanisms that are inadequately suited for AI applications, suggesting alternatives like value-based reimbursement.¹²⁶⁻¹²⁸

There are ways to balance AI's benefits while mitigating its potential risks. AI is a new technology, and physicians will require education and training to understand how AI models work, where they can be applied, and what benefits and risks are associated with their use in clinical practice. Efforts to incorporate AI education into the curriculum are currently underway.¹⁸ To ensure generalizability of the models, AI must be applied to diverse and representative datasets and populations. This will also ensure equity in delivery of care.¹²⁹ Validation of AI algorithms in larger, nationally representative cohorts will also support better accuracy.^{18,20} Data security may be achieved with encryption and maximizing use of deidentified or coded patient information. Regulatory safeguards will also be necessary to protect patients, given considerations regarding whether patients should be aware of their data's being used in AI applications and specifically consent to such use.¹⁹ Costs can be mitigated by considering the potential health care savings resulting from timely diagnosis and intervention.¹³⁰ However, several hurdles to implementation and reliability remain to be addressed for AI systems that promise to reduce administrative costs.⁸⁴

In addition to health care providers, patients will be asked to embrace AI too. Their engagement as a key stakeholder will necessitate education to enable better understanding of what AI is, why its use is beneficial in health care settings, what benefits and risks are associated with it, and how it can improve patient outcomes and support personalized care. AI may facilitate better use of existing resources, minimize waste, and provide efficient, patient-centered care. By addressing AI-associated risks, its transformative potential in allergy and immunology can be fully realized, thereby benefiting patients, clinicians, and the broader health care system.^{3,18}

ROAD TOWARD IMPACT

Despite an exponential rise in publications about AI, its real-world impact in allergy and immunology remains limited. Recent years have witnessed increased efforts by policymakers to implement AI regulations and governance frameworks aimed at maximizing benefits while mitigating potential risks. Although this represents important and urgent steps forward, we contend that additional strategies are needed to bridge the persistent gap between research and clinical implementation in health care. We provide a perspective of a 6-point road map to focus AI development efforts toward increasing its clinical impact (Fig 4).

1. Prioritize impactful challenges and underutilized data

AI should address clinically significant gaps where current solutions are inadequate rather than optimizing already functional workflows. The goal should be to tackle challenges where traditional methods fall short. Recent bibliometric and review studies show that much of AI research in allergy and immunology targets low-impact or already-solved problems, such as adolescent asthma diagnosis.^{21,131} Tackling more difficult clinical problems, such as diagnosing asthma in a preschoolers, would provide more impact.

One promising direction is unlocking underutilized data sources, such as clinical notes, wearable sensor data, seasonal exposure, and multiomics datasets,^{19,132,133} aided by novel developments in the field of AI. Research should prioritize applications that leverage these untapped information streams. These can be combined with structured data sources in multimodal analysis approaches.

2. Define clinically relevant benchmarks

AI performance should be assessed against real-world clinical standards in addition to just controlled, idealized scenarios. Even among FDA-approved, AI-enabled medical devices, just 55.9% reported clinical performance metrics.¹⁵

Beyond standard metrics, studies must evaluate cost-effectiveness, scalability, and feasibility of implementation. The total cost of AI adoption—including development, integration, and operational expenses—must be explicitly accounted for. We encourage the development of standardized benchmarks to assess AI's clinical relevance within health care.¹³⁴⁻¹³⁶

3. Enforce rigorous validation and governance

Studies must move beyond single-cohort evaluations to large-scale, externally validated trials that reflect real-world patient populations. While more recent and AI-specific figures are missing, asthma prediction models up to 2017 showed a major risk of bias and a lack of use of external validation.^{137,138}

Although randomized clinical trials remain encouraged to assess the effectiveness of AI tooling,¹³⁹ they remain scarce, with only 86 reported in a 2024 scoping review.¹⁴⁰ Concurrently, proposals have been made to improve randomized clinical trials to better capture the clinical value of AI¹¹³ by integrating implementation success assessment alongside traditional efficacy

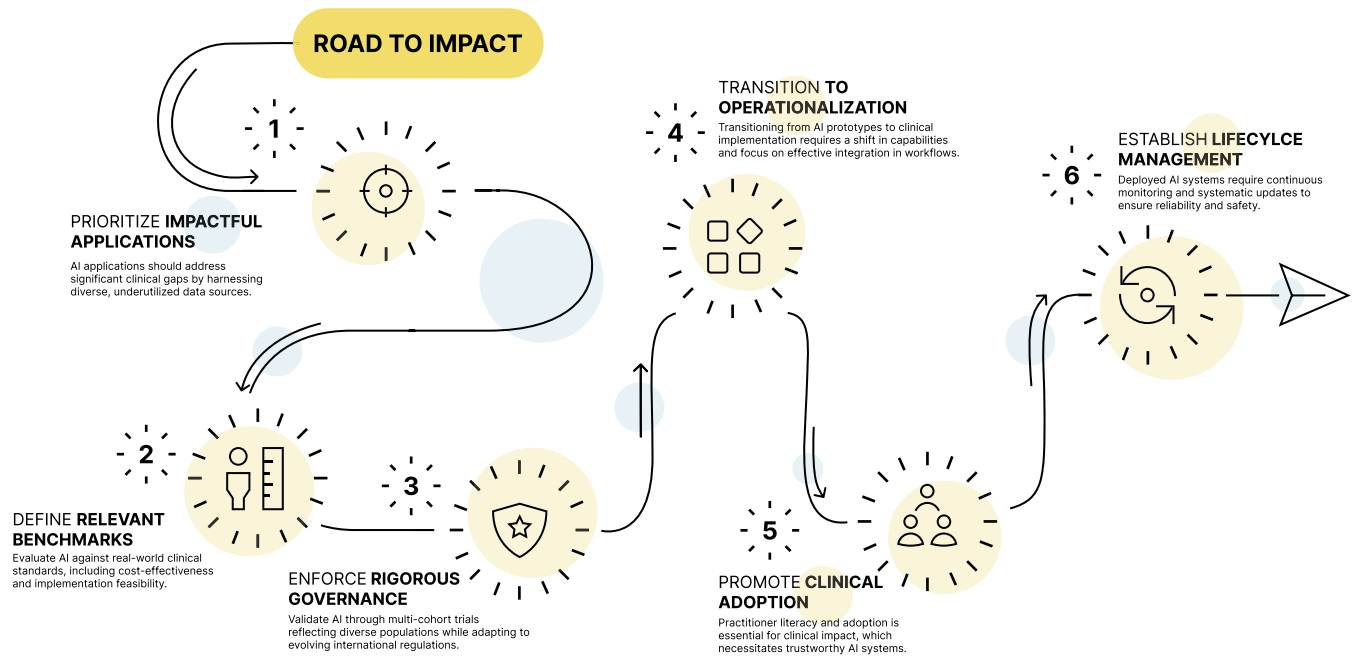


FIG 4. Road to impact of AI in allergy and immunology.

measures. Similarly, various studies showed that LLMs are being evaluated poorly and not representatively for true clinical performance.^{55,141,142}

Legal governance for AI is also evolving, especially with the European Union's AI Act being put into effect,¹⁴³ representing a significant step toward responsible AI deployment in health care. In contrast, the United States approaches AI regulations differently.^{144,145}

4. Move from experimentation to operationalization and clinical implementation

Realizing the full potential of AI in clinical practice necessitates the development of robust, production-ready applications. However, the transition from a research prototype to an operationalized system demands substantial resources, diverse technical expertise, and cross-functional collaboration among specialists.¹⁴⁶⁻¹⁴⁸ To bridge this gap, we strongly advocate for strategic partnerships between researchers and AI practitioners. Implementation frameworks can guide efficient integration of AI into existing workflows, such as the SALIENT framework,¹⁴⁹ building on TRIPOD-AI, DECIDE-AI, CONSORT-AI, and TE-HAI.¹⁵⁰⁻¹⁵³ This framework introduces essential components for production-level AI systems, reflecting industry best practices to move beyond proof of concept. In the short and mid term, setups with humans in the loop are still required in clinical settings.

However, SALIENT requires adaptation for allergy and immunology applications: it lacks guidance for the continuous model updating needed for dynamic allergen sensitization patterns and evolving phenotypes, it provides insufficient regulatory pathways for component-resolved diagnostics under new AI regulations, and it overlooks outpatient-centered delivery of care. These limitations are particularly relevant given our field's emphasis on pediatric populations, home monitoring systems,

and adaptive AI requirements for seasonal variations and emerging allergens.

5. Promote adoption through trustworthy AI

Health care leaders have raised concerns about the adaptability and trialability of AI,¹⁵⁴ emphasizing the need for user-centered implementation approaches and adoption. While AI adoption varies across medical disciplines, health care generally demonstrates lower AI maturity compared to other industries.¹⁵⁵ Systematic adoption frameworks, such as NASSS,¹⁵⁶ can guide adoption strategies. As AI systems increase in complexity, comprehensive explanations of all internal mechanisms may become infeasible; however, this parallels other accepted medical technologies. Clinicians do not need to fully understand the internal workings of a magnetic resonance imaging machine to utilize it effectively.

Such frameworks do, however, lack AI-specific guidance for challenges unique to allergy and immunology, including algorithmic bias in diverse phenotypes, hallucination risk, performance drift as allergen sensitization patterns and their relation to clinical outcomes evolve over time, and data-quality issues from heterogeneous populations in diverse environments. Additionally, the framework overlooks the fact that allergy practitioners may have lower AI exposure than technology-driven specialties like radiology, thus creating barriers to adoption that require field-specific change management strategies.

6. Ensure postdeployment monitoring and systematic updating frameworks

After clinical adoption, AI systems require life cycle management encompassing both performance monitoring and systematic updating protocols. Monitoring frameworks should continue to evaluate diagnostic accuracy, cost-effectiveness, and data-drift

detection, incorporating clinician feedback mechanisms for incorrect decisions.¹⁵⁷ When monitoring identifies performance degradation, updating mechanisms become essential.¹⁵⁸ Just as physicians require continuing medical education, AI systems necessitate regular updates incorporating emerging clinical evidence, revised treatment guidelines, and adaptations to evolving operational contexts. Health care professionals similarly require ongoing training to maintain proficiency with these technologies. Although the FDA provides some guidance for life cycle management,¹⁵⁸⁻¹⁶⁰ only a fraction of FDA-approved devices has reported retraining since market entry.¹⁶¹

As AI applications for allergy and immunology emerge, robust monitoring and updating of AI systems becomes particularly crucial, given the heterogeneity of the field's patient populations and the complexity of allergic and immunologic conditions.

CONCLUSIONS

The most recent developments in AI—LLMs, multimodal learning, and AI agents—progress quickly and have already been applied in the field of allergy and immunology. The use of AI in allergy and immunology is increasing, with potential for integration in multiple areas including diagnosis, management, prevention, self-care, and research. Many research articles and proof-of-concept reports have been published, underscoring the technology's great potential. However, the translation of promise to tangible impact from AI in medicine is clearly a major challenge. We propose to shift focus to bridge the gap between research and clinical implementation. Therefore, we suggest a 6-point road map that focuses on impactful challenges, clinically relevant benchmarks, rigorous validation, operational implementation, trustworthy adoption, and life cycle management. By shifting research priorities toward real-world impact, AI can move beyond theoretical promise and drive tangible improvements in allergy and immunology care.

DISCLOSURE STATEMENT

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